

Multilayer Perceptron for Breast Cancer Classification

Aim

To implement a Multilayer Perceptron (MLP) classifier for breast cancer classification using:

1. a machine-learning library, and
2. a from-scratch implementation using NumPy, forward propagation, and backpropagation.

Dataset

Use the Breast Cancer Wisconsin Diagnostic Dataset available through `sklearn.datasets`. The dataset contains:

- Number of samples: 569
- Number of features: 30
- Number of classes: 2
- Target classes: Malignant and Benign

Part A: MLP Using a Machine-Learning Library

Task

Develop an MLP classifier to predict whether a breast tumor is malignant or benign.

Students must perform the following steps:

1. Load the Breast Cancer dataset.
2. Separate the features and target variable.
3. Perform appropriate feature scaling.
4. Divide the dataset into training and testing sets.
5. Construct an MLP classifier containing:
 - Input layer
 - One or more hidden layers
 - Output layer
6. Train the MLP using the training data.
7. Predict the classes of the test data.
8. Evaluate the trained model.
9. Display the classification results.

Experimental Study

Students must investigate the effect of changing the following parameters:

No.	Parameter to Investigate
1	Number of hidden layers
2	Number of neurons in each hidden layer
3	Learning rate
4	Number of training iterations/epochs
5	Activation function

Table 1: MLP parameters for experimental study

Students should change one parameter at a time while keeping the other parameters fixed and observe the effect on model performance.

Questions

1. What happens to the model when the number of neurons in the hidden layer is increased?
2. What happens when the learning rate is very small?
3. What happens when the learning rate is very large?
4. Compare an MLP with one hidden layer and an MLP with two hidden layers.
5. Why is a nonlinear activation function required in the hidden layers?
6. What happens if all the layers use a linear activation function?
7. Why can increasing the number of hidden neurons sometimes lead to overfitting?

Part B: MLP From Scratch

Task

Implement an MLP from scratch for the same Breast Cancer Wisconsin Diagnostic Dataset.

The implementation must not use an existing MLP or neural-network implementation from a machine-learning library.

Students may use *NumPy* for matrix operations and numerical computations.

The implementation must include the following components:

1. Weight initialization
2. Bias initialization
3. Forward propagation

4. Activation function
5. Loss/error calculation
6. Backpropagation
7. Weight update
8. Bias update
9. Iterative training
10. Prediction